

THE LEDGER

VOL. CXLII · NO. 214

THURSDAY

LATE EDITION · A1

FAIR LENDING

Bank Denied the Loan It Approved Next Door

Reporters matched hundreds of near-identical mortgage applications in the lender's own public filings. Same income band, same block, same collateral — opposite answers.

BY A STAFF REPORTER

The comparison took a weekend and a laptop. Every record used in the analysis was published by the bank itself, as federal law requires.

In one pair, two applications for homes on the same block, at the same appraised value and in the same income band, were separated by four-tenths of a point of loan-to-value. One was approved. The other was denied for collateral.

A spokesperson said the lender's standards are "consistently applied," and declined to discuss individual files. Two state attorneys general have since asked for them.

You don't want to be that company.

Nothing in the story required access the public doesn't have. The only advantage the bank could have had was reading its own file first.

 **LIKEWISE**

THE LEDGER

VOL. CXLII · NO. 219

TUESDAY

OPINION · A18

EDITORIAL

A Lender That Could Show Its Work

Meridian Trust closed its fair-lending review with no adverse findings — and, unusually, with a record of what it had already fixed.

THE LEDGER'S VIEW

Passing a governance and ethics review is not supposed to be news. That it is says more about the industry than about this bank.

Meridian did something its peers do not: it read its own published file the way an examiner would, compared applications against each other, and found fourteen denials whose stated reasons did not survive the comparison. It reversed eleven

and rewrote the standard behind the rest — months before the review opened.

Its competitors had the same data available to them. What they lacked was the willingness to look.

You want to be this company.

Same public record, same method. The only difference is who read it first — and whether the fixes were already in the file.

Likewise reads a lender's own public record and names the two applications an examiner would put side by side — one approved, one denied, on the same facts.

Matched-pair fair-lending detection across all 13.5M published U.S. mortgage decisions.

The plan

01	Company purpose	Two identical applications, one answer	06	Competition	Crowded, and all statistical
02	Problem	Nobody reads their own file	07	Product	Four steps, one of them new
03	Solution	The matched pair, named	08	Business model	Free scan, then SaaS
04	Why now	The record is public	09	Team	Four roles, fourteen weeks
05	Market size	4,782 filers, \$86M/yr	10	Financials	Pre-revenue, and specific about it

Structure follows Sequoia Capital's ["Writing a Business Plan"](#).

Did you deny one application while approving a materially identical one?

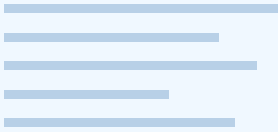
Answered today by hand, on a sample, once a cycle.

5 · Likewise — confidential

 LIKewise

+

She has everything except a file

 3.8pts A DISPARITY	 A REPORT	 40,000 FILES	 WHICH TWO
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The first three describe a portfolio. Only the fourth survives an exam.

How it works today



We name the pair

APPLICATION A

APPROVED

Loan purpose	Home purchase
Occupancy	Principal residence
Lien	First
Income band	\$75–100K
Property value	\$285,000
Census tract	34013-0072
Loan-to-value	94.2%

APPLICATION B

DENIED – COLLATERAL

Loan purpose	Home purchase
Occupancy	Principal residence
Lien	First
Income band	\$75–100K
Property value	\$285,000
Census tract	34013-0072
Loan-to-value	94.6%

Collateral does not distinguish these files.

Comparative file review is a documented step in the Interagency Fair Lending Examination Procedures.

Examiners already trust it. They just do it by hand, on samples, once a cycle. We run it exhaustively, and continuously.

The record is already public

13.5M

mortgage records
published for 2025

4,782

institutions that filed

2.1M

denials a year, each with a
recorded reason

\$0

what the data costs



Your file is already open. You're the last one reading it.

A state attorney general, a journalist, an advocacy group or a plaintiff's firm can run this analysis on you today. DOJ's redlining initiative recovered \$107M+ before the department began unwinding consent orders in 2025-26.

11 · Likewise — confidential



Market size

TIER	SIZE	BASIS
SOM	\$36M/yr	~2,000 mid-tier filers × ~\$18K
SAM	\$86M/yr	All 4,782 HMDA filers
TAM	\$1.4B → \$4.2B	Fair-lending compliance software, ~12.7% CAGR

Competition

EVERYONE ELSE

"Is there a disparity?"

A coefficient the bank litigates

Regression on protected class

LIKEWISE

"Which two files, and why did they differ?"

An exhibit the bank fixes

Exhaustive pairwise matched-pair detection

WOLTERS KLUWER

FISERV

SAS

NCONTRACTS

ASURITY RISKEEXEC

COMPLIANCETECH

TRUPOINT

RATA

Eight vendors in this category. None of them names the pair.

Product

01

Block & match

Pairwise similarity over mixed features, at portfolio scale.

02

Rerank

A learned model decides what materially identical means across threshold features.

03

Test the reason

Does the stated denial reason hold in the comparison file? Nobody else does this step.

04

Rank for review

A queue for a human, ordered by how weakly the reason survives.

IN: 13.5M PUBLIC RECORDS

OUT: RANKED PAIRS

Evaluation

WHAT WE MEASURE

Match quality

Label quality

Reason-presence accuracy

Calibration

Baselines

HOW

F1 on 44,691 human-labelled duplicate pairs

Cohen's κ , published

300 hand-labelled pairs

Brier score and reliability curve

TF-IDF and embedding retrieval

Wedge, price, moat

Free scan

We run a prospect's own public file before they sign anything. The data being public is the wedge.

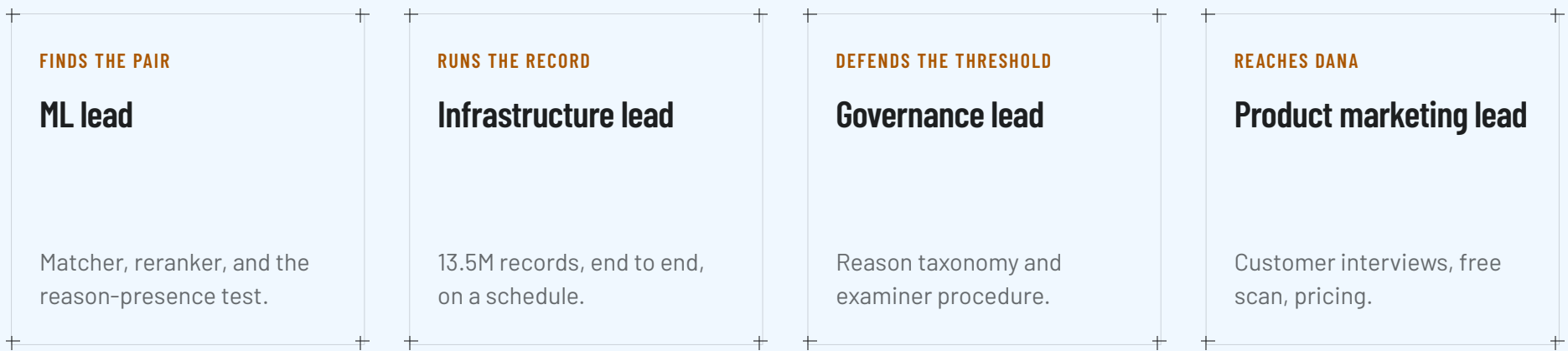
\$12-30K a year

SaaS, priced by institution size. The price point is a hypothesis we are validating in customer interviews.

The moat

The validated matcher and the labelled evaluation set — not data access.

Who owns what



Fourteen weeks in: matcher with published F1 and κ , calibrated outcome model, dashboard over real HMDA data.

Financials

\$12–30K

Price per institution per year — a hypothesis, in validation now

\$0

Marginal cost of a prospect's first scan — the data is free

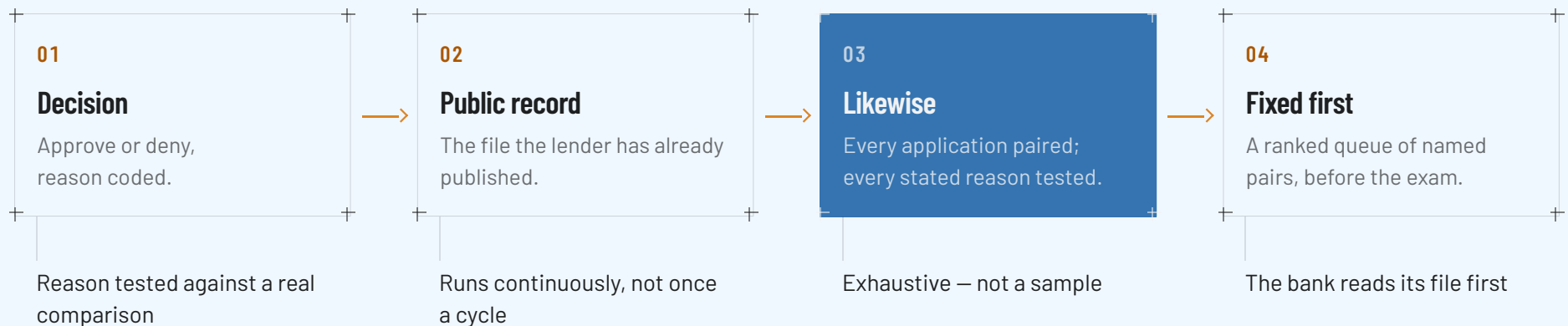
2,000

Mid-tier filers we can reach with the free scan, unaided

What we do not have yet: a signed contract, a renewal, or a sales cycle we have measured. Those are the next fourteen weeks.

WITH LIKEWISE · SAME CHAIN, ONE STEP INSERTED

How it works with Likewise



TWO IDENTICAL APPLICATIONS

One answer

APPLICATION A	APPROVED	APPLICATION B	DENIED
Income band	\$75-100K	Income band	\$75-100K
Property value	\$285,000	Property value	\$285,000
Census tract	34013-0072	Census tract	34013-0072
Loan-to-value	94.2%	Loan-to-value	94.6%

LIKewise

Reason does not survive

Denied for collateral, on the same collateral. Ranked first for review.

The distinguish engine

What it decides

Which feature differences are material

Which are within normal underwriting noise

Whether a stated reason survives the comparison

How much of that judgement is contestable

Why it's a research question

No statute or rule defines materiality numerically

94.2% against 94.6% is the entire case

Loose thresholds indict everyone; tight ones, no one

The threshold is policy, not a hyperparameter

What we'd publish, not sell

RQ 01

What makes two decisions comparable?

A contestable, published definition of feature-wise materiality for automated decision review — auditable by someone who disagrees with it.

→ [MATERIALITY SPEC, OPEN](#)

RQ 02

Can a stated reason be tested?

Reason-validity as a measurable property of a decision record, with precision and recall reported against human adjudication.

→ [REASON TAXONOMY + BENCHMARK](#)

RQ 03

Whose ground truth counts?

Inter-annotator agreement as a governance artifact: what a model may claim about fairness given the k of the labels beneath it.

→ [LABELLED EVALUATION SET](#)

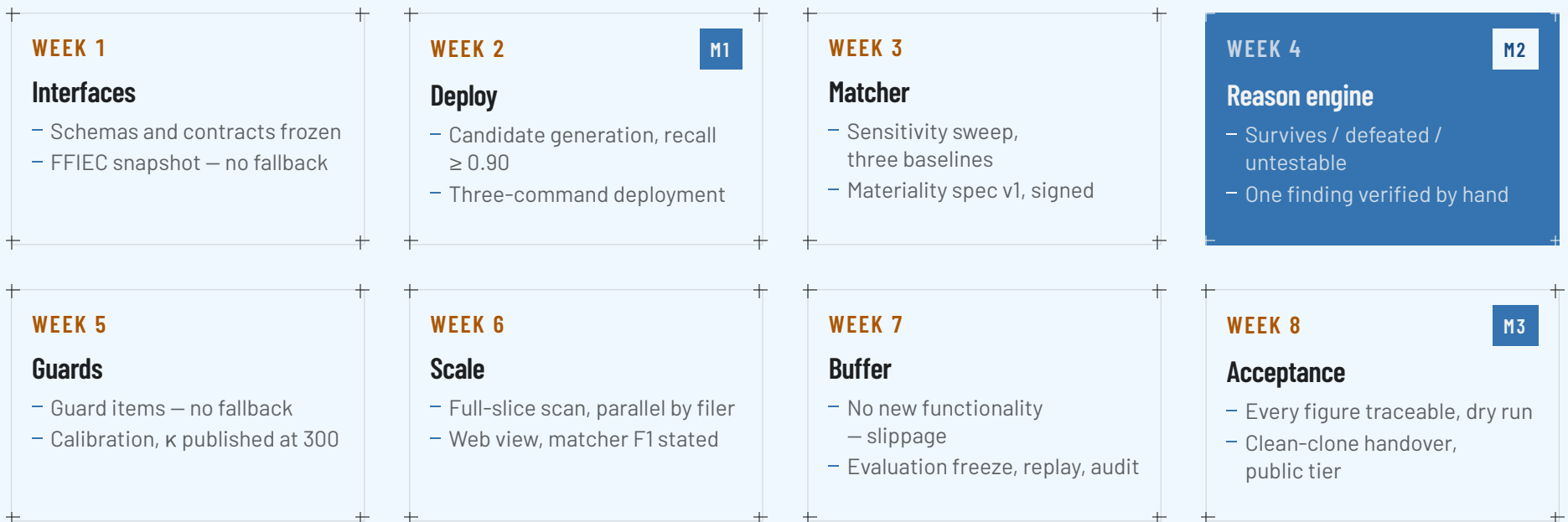
RQ 04

Can you audit without protected attributes?

Outcome-based matched-pair auditing transferred to hiring, insurance and benefits decisions, where class data is unavailable by design.

→ [AUDIT HARNESS, NON-COMMERCIAL](#)

Eight weeks to v1



Where the data comes from

SOURCE	WHAT WE TAKE FROM IT	LINK
HMDA Snapshot dataset	Every filer, every application, one vintage	ffiec.cfpb.gov/.../snapshot-national-loan-level-dataset
Modified LAR, per filer	One prospect's own portfolio — the free scan	ffiec.cfpb.gov/data-publication/modified-lar
Data Browser and API	Slice selection and cross-checks	ffiec.cfpb.gov/data-browser
Filing Instructions Guide	Schema and the denial reason codes	ffiec.cfpb.gov/documentation
FFIEC census and income	Income bins and tract characteristics	ffiec.gov/censusapp.htm
Regulation C, 12 CFR 1003	What the record contains, what privacy strips	consumerfinance.gov/rules-policy/regulations/1003
Fair lending exam procedures	The matched-pair standard we automate	ffiec.gov/PDF/fairlend.pdf
Pre-2017 HMDA products	Back years, for trend claims we source	ffiec.gov/hmda/hmdaproducts.htm

No licence, no vendor, no acquisition cost. Every figure in this deck traces to one of these eight files.

The headline is a negative result, and it is the strongest thing here.

13,543,606 records. 4,782 filers. 2,883 scans, one filer at a time — because the question is about a filing against itself.

RESULT • NATIONAL SCALE, FIRST RUN

372 findings from 2 million denials

8,853,455

Records in scope, across
2,883 filer scans

2,013,872

Denials examined against
the filer's own approvals

372

Findings surviving the
power floor and k-
anonymity

3 of 1,284

Filers sitting above their
own permutation null

The test is signed so positive means denials look **better** than chance on their cited dimension. The national distribution runs the other way — median $z = -1.18$, 81% of filers negative.

 **LIKEWISE**

8.85M to 372, and every step names what it drops

STEP	COUNT	WHAT THE STEP IS
Records in scope	8,853,455	The snapshot, one filer at a time
Denials in scope	2,013,872	The population under test
Citing a reason the record can carry	720,827	35.8% — the rest name credit history or documents
With a matched comparator	357,285	Same coarsened cell, approved, not worse
With a dominating comparator	495	Published values coarse enough to separate the pair
Findings after the power floor	372	13,687 more were removed by k-anonymity

[230,237](#) denials cite only reasons the file has no column for. [73,243](#) are testable in principle, unresolvable in practice. Disclosure control, not statistics, removes most of what the engine finds.

METHOD · EVERY STATISTIC IMPLEMENTED TWICE

Six models, none of them the obvious one

PROBLEM

Values are intervals, not points

Median cell holds 3 records

Ties everywhere

Is a finding rate meaningful?

Not randomised

Pairs are dependent

MODEL USED

Interval dominance

van Elteren / CMH

Midrank variance

Exact closed-form null

Rosenbaum Γ^*

Cluster bootstrap

WHY NOT THE OBVIOUS CHOICE

A score would invent an ordering the file cannot support

No within-cell test can reject: smallest p in a cell of 3 is 1/3

The textbook form would make the test silently conservative

A raw rate is mostly geometry; the closed form removes the seed

You cannot rule out unmeasured confounding — you can price it

Resampling rows treats dependent observations as independent

10 of 10 checks agree with SciPy / statsmodels — and found a real overflow defect in the release gate that no unit test could reach.

Only simulation says whether the design works

01 · Calibration

12 of 12 configurations calibrated, none anti-conservative

Four publication regimes × three filing sizes. Rejection at $\alpha = 0.05$: 0.047 / 0.054 / 0.048 / 0.050.

02 · Detectable effect

MDE falls as $1/\sqrt{\text{cells}}$ – 1.42, 0.84, 0.46

At 100, 300 and 1,000 cells. A rank test is largely indifferent to binning: it preserves order even where it destroys distance.

03 · Attainability

No cell can reach $q = 0.05$

Best attainable p in any simulated cell is 1/8. Per-finding p-values and FDR were implemented, found unattainable, and deleted.

THE DESIGN POOLS ACROSS CELLS

04 · Geometry

Under a true null the rule still fires on ~35% of cells

Cluster bootstrap covers 0.932 of nominal 0.95; the naive row bootstrap covers 0.858 while being 22% narrower.

PUBLISH THE DEPARTURE, NEVER THE RATE

NATIONAL RESULT · 1,270 P-VALUES COMBINED BY FISHER

Almost every lender sits below its own null

-1.18

Median filer z . $\chi^2 = 2,325$
on 2,540 df; lower tail p
= 0.0010

420 vs 9

Filers below -1.645, against
nine above +1.645

55%

1,567 of 2,837 filers
contribute no testable cell
at all

24

Filers producing any finding
– 366 of 372 from 10k+ filers

One defect no test could have found: four filings (~190,000 records) refused at load because six records nationally carry `property_value = 2,147,483,647` – INT32_MAX, undocumented as a sentinel. The rule is right; the diagnosis is wrong.

What this establishes, and what it does not

Established

The estimators are correct. 10/10 agree with independent implementations.

The test is calibrated. 12/12 configurations, never anti-conservative.

The constraint is arithmetic, not power. 55% of filers have nothing testable.

The record is consistent with its own reasons. Median $z = -1.18$; 3 filers above null.

Not established — say it out loud

Not a finding about any lender. The output is a review queue, never a determination.

A negative z is not a clean bill of health. Credit score and documentation are absent from the file.

Nothing was pre-registered. Every scan carries analysis_status: exploratory.

Comparability is bought with volume. The product serves best the filers who need it least.

NEXT RUN · EFFECTS MEASURED, NOT GUESSED

What would change the conclusions

CHANGE

Block on MSA-or-state instead of county

Ordinal comparison over published DTI bands

One-sided control tolerances

Add 2147483647 to the numeric sentinels

MEASURED EFFECT

Powered fraction 2.7% → 5.7%

Matched pairs comparable 5.1% → 99.6%

Powered fraction → 17.6% (6.5×) — exchangeability must be restated first

Recovers 4 filings, ~190,000 records

The result was a method that can say, with a number, when the record cannot answer the question.

This run is the baseline every one of those changes has to beat. FFIEC 2025 snapshot · engine v0.14.0 · materiality spec 1.5.0 · 2,553,576 candidate pairs · median scan 2.5 s.